



< RAMRI
ReMynd Authentic Mathematical Reasoning Interview

8 approved samples



1. Upload

2. Samples

3. Sample Bank

4. Choice Sets

5. Interview

Scoring & Report

Rate each reasoning domain 0–4 or mark as Not Observed. Evidence strength reflects confidence in the rating.

RAMRI Disclaimer: RAMRI is a structured qualitative and criterion-referenced reasoning interview. Results must not be represented as standardized scores, age equivalents, grade equivalents, or diagnostic conclusions.

Mathematical Reasoning Domains

Conceptual Understanding

Developing with moderate support

0

1

2

3

4

NO

Evidence strength...

Supporting evidence...

Strategy Awareness

Emerging with substantial support

0

1

2

3

4

NO

Evidence strength...

Supporting evidence...

Procedural Reasoning

Developing with moderate support

0

1

2

3

4

NO

Evidence strength...

Supporting evidence...

Mathematical Communication

Demonstrated with limited support

0

1

2

3

4

NO

Evidence strength...

Supporting evidence...

Error Awareness

Developing with moderate support

0 1 **2** 3 4 NO

Evidence strength... ▼

Supporting evidence...

Verification and Reasonableness

Emerging with substantial support

0 **1** 2 3 4 NO

Evidence strength... ▼

Supporting evidence...

Strategy Flexibility

Emerging with substantial support

0 **1** 2 3 4 NO

Evidence strength... ▼

Supporting evidence...

Transfer

Emerging with substantial support

0 **1** 2 3 4 NO

Evidence strength... ▼

Supporting evidence...

Metacognition

Developing with moderate support

0 1 **2** 3 4 NO

Evidence strength... ▼

Supporting evidence...

Independence

Developing with moderate support

0 1 **2** 3 4 NO

Evidence strength... ▼

Supporting evidence...

✓ Save Ratings



Background Evidence (214 items)

— not selected for interview; included in AI report context ▼

Examiner Documents

PDFs or documents you attached — included in AI report generation.

📎 Attach file

No examiner documents attached yet.

Clinical Report

Status: **draft**

🔄 Regenerate Report

⚠️ AI-generated report — human review and approval required before release.

Assessment Context

The RAMRI (Reasoning About Mathematical Reasoning Interview) is a criterion-referenced qualitative interview designed to understand a student's mathematical thinking processes. This session involved a structured discussion with Cayline Yining Cheng about her own previously completed math work. A total of seven work samples were selected and discussed in detail during the interview, with one additional approved sample available for written work.

Participation & Emotional Presentation

Cayline exhibited varying levels of engagement and confidence throughout the assessment. She demonstrated confidence and perceived ease on tasks she found straightforward, such as identifying vertices, stating, "This is very easy. You just need to count them" (Sample 1). On problems where she possessed a clear strategy, such as single-digit addition, she was actively engaged, self-correcting her response from 19 to 17 during the interview after realizing she had miscounted (Sample 2). She also articulated her reasoning for preferring a number chart for larger addition problems, explaining it "allowed her to count forward more efficiently" (Sample 5). However, when confronted with tasks where her conceptual understanding was weak or she lacked a defined strategy, such as skip counting or missing minuend problems, Cayline's engagement significantly decreased. She frequently provided "No response provided" to follow-up questions (Samples 3, 6, 7) or directly stated, "she did not know how to solve the problem" (Samples 6, 7), indicating a perceived difficulty and subsequent disengagement. Her metacognitive awareness was also variable; she insightfully noted that "counting the circles could be difficult because she might miscount them" (Sample 2) but did not comment on the difficulty or her thinking process for other incorrect answers (Samples 4,

Mathematical Reasoning Profile

understanding, strategy articulation, or error awareness are required. Her **Conceptual Understanding** is rated 2/4. She demonstrated understanding of geometric terms by correctly circling three vertices on a triangle (Sample 1) and recognized addition problems (Sample 2: "contained an addition equation"). However, she was "unable to explain what the task required or demonstrate an understanding of counting by twos" (Sample 3) and lacked conceptual insight into missing minuend problems, stating she "did not know how to solve the problem" (Samples 6, 7). Her **Strategy Awareness** is rated 1/4. Cayline could describe her counting strategies for vertices (Sample 1) and drawing circles for single-digit addition (Sample 2), and she showed adaptation by using a number chart for larger addition (Sample 5). Yet, she was "unable to describe how she generated the sequence or explain the strategy she used" for skip counting (Sample 3) and provided "No response provided" for strategy questions on missing minuend problems (Samples 6, 7). In terms of **Procedural Reasoning** (2/4), Cayline correctly identified vertices (Sample 1) and successfully used a number chart to solve $12+6=18$ during the interview (Sample 5). However, her initial written answers for $9+8$ were incorrect ("20", Sample 2) and for $12+6$ was "6" (Sample 5). She also showed a procedural breakdown in skip counting, writing "11, 12, 13, 14, 15" (Sample 3) and failed to complete missing minuend problems despite partial written work (Samples 6, 7). Her **Mathematical Communication** is rated 3/4. Cayline clearly articulated steps for simple tasks like counting vertices (Sample 1) and drawing circles (Sample 2). However, communication became limited, often resulting in "No response provided," when conceptual understanding was lacking (Samples 3, 6, 7). **Error Awareness** is rated 2/4. Cayline self-corrected for $9+8$ when prompted to demonstrate (Sample 2), but she "did not identify any

Per-Domain Findings

The interview revealed specific patterns in Cayline's reasoning across different mathematical domains.

Geometry: For **Vertices Identification** (Sample 1), Cayline's written work showed a drawing of a triangle with three circled vertices, indicating a correct answer. During the interview, she explained, "This is very easy. You just need to count them" and that "A triangle has three vertices, so it's very simple." For **Shape Properties (Sides and Vertices)** (Sample 4), her written work indicated she marked the square and trapezoid but did not mark the rhombus, and she only selected two shapes. Examiner notes confirm she "did not select the rhombus which also has 4 sides and 4 vertices" and "did not notice that the question required her to 'choose three that apply.'" She verbally confirmed counting sides and vertices but "did not identify any errors in her response."

Addition Reasoning: In **Single-digit addition** (Sample 2) for $9 + 8$, Cayline's written answer was "20." During the interview, she explained her strategy of drawing and counting circles. Upon demonstration, she "realised that she had miscounted the circles" and "independently corrected her response" to 17. For **Addition within 20** (Sample 5) for $12 + 6$, her initial written answer was "6." However, during the interview, she described using a 1–100 number chart, "located 12 on the chart, counted forward six numbers, arrived at 18, and recorded 18 as her answer." She "was unable to identify any errors" in her original written answer.

Number Sense: For **Skip Counting by 2s** (Sample 3), Cayline's written answer showed "11, 12, 13, 14, 15" following "2, 4, 6, 8, 10." Examiner notes highlight a shift to counting by 1s. She explained, "I just remembered it: 2, 4, 6, 8, 10, 11, 12, 13, 14, 15" and was "unable to explain what the task required or demonstrate an understanding of counting by twos."

Performance vs. Reasoning

Cayline's performance on written work samples sometimes diverged from the reasoning she articulated during the interview. For the addition problem ' $9 + 8 = \underline{\quad}$ ', her initial written answer of '20' (Sample 2) indicated an error, yet when asked to demonstrate, she accurately applied her strategy of drawing circles, identified her miscount, and 'independently corrected her response' to 17. This suggests that while initial execution can be flawed, her underlying

Effect of Assessment Conditions

The RAMRI's qualitative interview format, using Cayline's own work, appeared to mitigate anxiety in certain contexts and support engagement. On tasks Cayline perceived as 'very easy' (Sample 1), she demonstrated confidence and willingness to articulate her simple strategies. The opportunity to 'show me what you did' allowed her to actively engage in self-correction for the 9+8 problem, leading to an independently corrected answer of 17 (Sample 2).

Domain Coverage

This assessment explored Cayline's mathematical reasoning across the domains of Geometry, Addition Reasoning, Number Sense, and Subtraction Reasoning. In Geometry, she demonstrated an understanding of vertices (Sample 1) but struggled with attention to detail in problem instructions for shape properties (Sample 4). For Addition Reasoning, she possessed strategies for single-digit (Sample 2) and two-digit addition (Sample 5) but showed initial

Transferable Reasoning Strategies

Cayline demonstrates several reasoning strengths that are likely to support her learning in unassessed domains when she encounters them. Firstly, her **metacognitive awareness of procedural difficulty** is a strength; she noted that 'counting the circles could be difficult because she might miscount them' for addition (Sample 2). This self-awareness of potential pitfalls in her own methods is transferable to complex calculations in domains like fractions or

Strengths Narrative

Cayline demonstrates several commendable strengths in her mathematical reasoning, particularly in her approach to familiar and concrete tasks. She exhibits a clear conceptual understanding of basic geometric properties, as evidenced by her correct identification and circling of all three vertices on a triangle, confidently stating, "This is very easy. You just need to count them" (Sample 1). Her ability to articulate her chosen strategies is also notable; she clearly explained her method of drawing and counting circles for single-digit addition problems

Strengths

Conceptual understanding of vertices: demonstrated by correctly circling three vertices on a triangle (Sample 1).

Ability to articulate concrete strategies: explained solving vertex identification by counting (Sample 1) and single-digit addition by drawing and counting circles (Sample 2).

Strategic adaptation: chose to use a 1-100 number chart for two-digit addition for efficiency (Sample 5).

Capacity for self-correction: independently corrected her calculation for $9+8$ from 19 to 17 when demonstrating her process (Sample 2).

Metacognitive awareness of procedural difficulty: noted that 'counting the circles could be difficult because she might miscount them' for addition (Sample 2).

Recognition of problem type: identified tasks as addition problems due to equation structure (Sample 2, 5).

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Areas for Development

Conceptual understanding of skip counting: 'unable to explain what the task required or demonstrate an understanding of counting by twos' (Sample 3).

Conceptual understanding of inverse operations/missing minuend: stated she 'did not know how to solve the problem' and 'could only see the numbers 9 and 4 in the equation' for ' $___ - 9 = 4$ ' (Sample 6) and ' $___ - 6 = 9$ ' (Sample 7).

Strategy development and articulation for complex tasks: 'unable to describe how she generated the sequence or explain the strategy she used' for skip counting (Sample 3) and provided 'No response provided' for strategies on missing minuend problems (Samples 6, 7).

Attention to detail in problem instructions: 'did not notice that the question required her to 'choose three that apply'' for shape properties (Sample 4).

Independent error identification and verification: 'did not identify any errors' for shape properties (Sample 4) and was 'unable to describe a strategy for checking' $9+8$ (Sample 2), relying only on recounting (Sample 1, 5).

Metacognitive awareness of gaps: provided 'No response provided' for metacognition questions when unsure (Samples 3, 6, 7) and 'did not comment on the difficulty of the task or describe her thinking process' for other incorrect answers (Samples 4, 5).

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Strategies for School / Classroom Teachers

Explicit instruction on conceptual understanding for counting patterns: Provide structured lessons on the meaning behind skip counting (e.g., counting by 2s means adding 2 each time) using manipulatives or visual aids, addressing Cayline's statement models (e.g., part-part-whole mats, balance scales) to illustrate missing addend/minuend concepts, linking subtraction to addition, to address Cayline's lack of strategy and conceptual understanding for ' $___ - 9 = 4$ ' (Sample 6) and ' $___ - 6 = 9$ ' (Samples 6, 7).

Practice highlighting keywords and identifying the number of required responses in multi-part questions (e.g., 'Choose three that apply'), directly addressing Cayline's oversight in Sample 4 where she 'did not notice that the question required her to 'choose three that apply''.

Encouraging verbalization of mathematical thinking: Implement 'think-aloud' protocols where students explain their strategies and reasoning aloud to their peers or the teacher, building on Cayline's ability to articulate strategies for simpler tasks (Samples 3, 6, 7).

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Strategies for Parents / Home Support

Counting games with real-world objects: Practice skip counting by 2s, 5s, or 10s with everyday items like pairs of socks, coins, or steps, helping Cayline build a concrete understanding of 'counting by twos' beyond rote memorization, as she stated, 'I just

'What's Missing?' number puzzles: Engage in simple games or puzzles that involve finding a missing number in addition or subtraction sentences (e.g., ' $5 + ___ = 8$ ' or ' $___ - 2 = 6$ ') using small objects to count, to develop early conceptual understanding of

'Spot the Difference' in math problems: Review homework together and practice rereading instructions, asking, 'Is there anything else the problem is asking for?' to develop attention to detail, specifically addressing the issue in Sample 4 where she

Analog clock practice: Use an analog clock at home to regularly practice telling time to the hour and minute, discussing the positions of the hands and their corresponding values (e.g., minute hand on 6 means 30 minutes, not 10), directly addressing the

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Strategies for Tutors / Learning Support

Conceptual foundation for inverse operations (Missing Minuend): The tutor should use concrete manipulatives (e.g., unifix cubes, counters) to model 'take away' and 'missing start' scenarios for subtraction, then explicitly demonstrate how addition can

Developing flexible skip counting strategies: Employ a number line or 100-chart to visually demonstrate skip counting by 2s, emphasizing the pattern of adding 2 each time, rather than solely rote memorization. Guide Cayline to verbalize the 'add 2' rule, a reasonable answer for $9+8=?$, using a different strategy (e.g., drawing circles then using a number line), or working backward. The aim is to build systematic verification that Cayline was 'unable to describe' (Sample 2) or 'unable to identify any errors' (Sample 5) independently.

Metacognitive questioning for self-reflection: Regularly prompt Cayline with questions like 'How did you know to do that?', 'What makes this problem tricky?', 'How is this problem similar to/different from one we did before?' and 'How could you check your

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Recommendations

Implement targeted interventions to strengthen foundational number sense, particularly conceptual understanding of skip counting and inverse operations.

Foster the development of explicit problem-solving strategies beyond simple counting, encouraging verbalization and justification of methods.

Provide opportunities for independent error identification and the development of systematic verification strategies.

Encourage attention to detail in problem instructions through guided practice.

Consider a re-assessment in 6-12 months to monitor progress in these identified areas.

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Limitations

This is not a standardised assessment.

The quality of conclusions depends on the authenticity and context of submitted work.

Previously completed work may have involved unrecorded assistance.

Transfer evidence is important when interpreting independent understanding.

Domains not represented in submitted work cannot be assessed — absence from this report does not indicate difficulty in those areas.



This report requires human review and approval by a qualified professional before release.



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